A couple of points about Wayne Myrvold's paper

Cian Dorr

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1. What does the epistemic concept of probability have to do with credence?

> As has often been pointed, out the word "probability" has been used to cover (at least) two distinct concepts, a fact that Hacking (1975) has expressed by saying that probability is Janus-faced. One concept, the *epistemic* concept, has to do with degrees of belief; epistemic probability is a measure of certainty and uncertainty. The other concept, which Hacking calls the *aleatory* concept, is objective and physical, having to do with properties of some chance set-up. In contemporary philosophical parlance, these are often distinguished as *credence* and *chance*. (20)

Authors like Bernouilli and Laplace are said to officially use 'probability' only for the epistemic concept, although they slip into the aleatory usage when talking about examples.

But *what*, exactly, does the epistemic use of "probability" have to do with degrees of belief? The equation of it with credence suggests that when interpreted epistemically, 'it is probable to such-and-such degree that P' means 'X is confident to such-and-such degree that P' for some contextually relevant X (such as the speaker).

But does 'probable' (or its less formal synonym 'likely') ever have this straightforwardly psychological meaning? I doubt it.

- Suggestion: 'it is likely that P' (on the epistemic reading) stands to 'X is confident that P' as 'it is certain that P' stands to 'X is certain that P'.
- Suggestion: the meaning of 'it is likely that P' (on the epistemic reading) includes a *normative* element that's not part of the meaning of 'X is confident that P'.
- More specific suggestion: 'it is likely to such-and-such degree that P' (on the epistemic reading) is equivalent to 'X *ought* to be confident to such-and-such degree that P', where X is some contextually salient individual or group (often the speaker), and the evaluative standard governing 'ought' is the sort of thing people get at with the phrase 'fit with one's evidence'.
- We'd better not think of this 'ought' as governed by any demanding 'ought implies can' principle — figuring out how likely something is (epistemically) can be calculationally very difficult.

2. How different are the "epistemic" and "aleatory" uses?

There are lots of ordinary uses of 'probable' and 'likely' which seem in some sense epistemic, but which obviously don't fit this analysis.

> I'm not sure how likely it is that Henry II ordered the death of Thomas à Becket, but if you want I can find out by looking on Wikipedia.
>
> Assuming you're right that the coin landed the same way every time, it's either very likely to be double-headed or very likely to be double-tailed.
>
> By taking this genetic test, you can find out how likely you are to be descended from Genghis Khan.

- Some suggest that these uses should be understood in terms of some notion of "available" evidence—roughly, how confident one ought to be if one had all the available evidence.
- I suggest instead that they should be understood in terms of *conditional* credence: in these settings, 'It is likely to such-and-such degree that P' means 'X ought to have such-and-such credence in P conditional on whatever is the true answer to Q', where Q is some contextually relevant *question* (understood as a set of pairwise inconsistent, jointly exhaustive propositions). The true answer to Q could be something quite specific, including one's exact readout on the genetic test, a bunch of facts about Genghis Khan, etc.

This makes it natural to think of the epistemic and aleatory uses as belonging to a continuum. Purely epistemic uses correspond to a trivial Q; aleatory uses correspond to a very detailed Q whose true answer includes the dynamical laws and lots of pretty detailed facts about the relevant "setup".

- In the book, Wayne shows (following Poincaré and others) how in many cases where it's natural to think in aleatory terms, we'll get the same answers to 'how likely?' questions for a wide range of different pretty-detailed values of Q.

3. How are credences related to judgments about epistemic probability?

Some remarks in the paper (echoing a common pattern in the literature, both historical and contemporary) seem to equate the claim that something has a certain probability of being the case with the claim that we *judge* it to have a certain probability of being the case (on an epistemic reading of 'probability').

> Keynes's diagnosis of the error inherent in the Principle of Indifferene is that it rests… on the assumption that, for any two alternatives, A and B, either A is more probable than B, or else B is more probable than A, or else they are equally probable. If this is assumed, then, in cases in which one neither judges A to be more probable than B, nor B more probable than A, there is no alternative but to judge them to be equiprobable.

This looks like it's conflating *not judging that A is more probable than B* with *judging that A is not more probable than B*.

- Even if *A is more probable than B*, *A is less probable than B*, and *A and B are equally probable* exhaust the possibilities, logically there are 7 possible consistent combinations of judgment someone might have about these, namely:

  (1) judging that A is more probable than B

  (2) judging that A is less probable than B

  (3) judging that A and B are equally probable

  (4) judging that A is not more probable than B, not judging that A is less probable than B, and not judging that A and B are equally probable

  (5) judging that A is not less probable than B, not judging that A is more probable than B, and not judging that A and B are equally probable.

  (6) judging that A and B are not equally probable, not judging that A is more probable than B, and not judging that A is less probable than B.

  (7) not judging that A is not more probable than B, not judging that A is less probable than B, and not judging that A and B are equally probable.

Keynes's claims are often (including in other work by Wayne) associated with the doctrine of *imprecise credence*, understood as involving states where a certain person X is neither more confident in A than B, less confident in A than B, nor equally confident in A than B (while still being at least as confident in A as in C and at least as confident in B as in D for some C and D).

But for these claims about *judgment* even to be relevant to the topic of credal imprecision, it seems that one would need to endorse claims such as the following:

> If X is more confident in A than in B, then X judges that A is more probable than B.
>
> If X is equally confident in A and B, then X judges that A and B are equally probable.

These seem very weird to me!

- First, judgment seems to be some kind of mental *act*, whereas confidence is an ongoing *state.*
- Second: if I'm right that the operative interpretation of 'probable' is something *normative* along the lines of 'ought to be confident', these seem to implausibly rule out a kind of epistemic akrasia (e.g. remaining very confident that your child is innocent even while judging that you should not be confident).
- Third: in a case where you don't make any outright normative judgments about what credence you should have, won't you often split your credence between differ-

ent hypotheses about how confident you ought to be? E.g. couldn't I end up 50% confident that P in a case where I judge I ought to be either 70% or 30% that P, but am 50% that I ought to be 70% and 50% that I ought to be 30%?